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Reframing remote sensing in Africanist archaeology: ethics, power, and participatory practice

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Introduction

Landscapes are not just physical spaces – they are archives of lived experience, memory, and meaning. Around the world, land is not simply used or inhabited; it is remembered, narrated, and cared for through oral tradition, ritual practice, and embodied forms of social memory. These forms of knowledge, transmitted across generations, structure how communities interpret environmental change, define sacredness, and govern land use. Remote sensing technologies, when applied without attention to these dimensions, risk flattening complex historical and cultural landscapes into anonymous data layers. Ethical practice, therefore, requires more than consent for image acquisition – it demands collaborative interpretation, critical engagement with place-based epistemologies, and an explicit reckoning with how knowledge production intersects with social memory, land rights, and community self-determination.

The use of remote sensing instruments has rapidly expanded our records of archaeological settlement patterns and human behaviour at the landscape scale. Aerial and spaceborne platforms like satellites, light-detection and ranging (LiDAR), and aerial photographs are useful for tackling issues of survey coverage and site identification (Casana, 2021; Chase et al., 2012; Lasaponara & Masini, 2012; Leisz, 2013; Osicki & Sjogren, 2005). As processing methods advance (e.g., machine learning and Artificial Intelligence (AI)), these tools offer effective ways to accelerate the discovery of new archaeological sites by increasing time efficiency in large-scale analyses, accessing regions where physical visitation is difficult or prohibited, and increasing our ability to see and detect evidence of anthropogenic landscape changes that are hidden by vegetation or otherwise invisible to the human eye (Bennett et al., 2014; Davis, 2021; Magnini & Bettineschi, 2019). With the power afforded by remote sensing technologies, we must also pay close attention to exploitation, consent, harm, and other ethical problems that emerge with such methods in relation to local communities.

Within archaeology, specifically, such ethical dilemmas have long been overlooked, particularly in the context of local, Indigenous, and descendant (LID) community interests, territorial rights, and sovereignty (Davis et al., 2021). One of the first

examples of engagement with ethical problems comes from Myers (2010), who examines the role of surveillance and power imbalances present when viewing landscapes from a bird's-eye perspective. In recent years, there has been an uptick in scholarly attention to such issues, with particular emphasis placed on power dynamics (Bennett et al., 2022; Davis & Sanger, 2021; Rączkowski, 2020; Sanger & Barnett, 2021).

Remotely sensed imagery, because of its ability to record information without being physically present in an area, holds considerable power in terms of perception, access, and narrative building, and these data can be used in ways that negatively impact many LID communities if research is not carefully designed (Bennett et al., 2022; Fox et al., 2006; Sanger & Barnett, 2021). Global-scale remote sensing datasets with extremely high levels of spatial and temporal resolution exacerbate issues of power and surveillance, which must be confronted head-on to ensure that research agendas are equitable and avoid the mistakes and injustices of the past.

Collaborative and critical remote sensing approaches have the capacity to engage with communities that have historically been excluded from past research. Such research agendas could help empower highly vulnerable populations who have been victims of displacement and disenfranchisement (Dunnavant, 2021; Dunnavant et al., 2023). For example, remote sensing can empower these communities with new data to advocate for environmental policies that integrate with their traditional livelihoods. Such data can also help to record and reclaim their history. However, these technologies can also afford outsiders unrestricted aerial access to locations that might otherwise be off limits, and this requires careful and extensive discussion with LID communities regarding consent and ethics (see Davis et al., 2021). Ultimately, transparency and consent are an essential part of ethical participatory archaeological practice (Atalay, 2012; Douglass et al., 2018; Lyons, 2013; Wadsworth et al., 2021a). In this chapter, we review the current state of aerial and spaceborne remote sensing methods within African archaeology and advocate for collaborative, participatory approaches that engage LID communities in the use of these technologies. We then provide a case study from our work on Madagascar in which we exemplify how remote sensing can be integrated into a community archaeology project while alleviating ethical issues.

Power dynamics and the ethics of remote sensing technology

While remote sensing investigations continue to increase in frequency, it is important to contextualise these methods in historical context and think critically about how their use has caused, and can continue to cause, problems and exacerbate injustices against marginalised communities. To begin with, the earliest aerial and spaceborne monitoring systems arose due to their military capabilities, particularly surveillance of adversarial territories (Parrington, 1983). Aerial spy planes and satellite imagery were crucial for both Russian and American intelligence agencies for espionage during the Cold War (Rosenthal, 1968).

With a history deeply rooted in the military industrial complex, these technologies were designed to avoid detection and, therefore, consent by those being observed (Gogarty & Hagger, 2008). While archaeologists and social scientists more broadly may not use these technologies with malicious intent to circumvent consent and community engagement, the potential consequences of using these datasets covering landscapes owned, stewarded, and used by a range of community stakeholders require thoughtful consideration (Davis et al., 2021).

Several researchers have thought about remote sensing in terms of its power dynamics (Davis et al., 2021; Myers, 2010; Rączkowski, 1999, 2020; Wadsworth et al., 2021a, 2021b). In particular, remote sensing can be thought of in terms of the panopticon (Bentham, 1791). A panopticon is an architectural style that was used in many European prisons in the eighteenth century, wherein a central viewing tower was positioned above a circular room of cells. Guards in the tower were not observable by inmates, and therefore forced prisoners to assume guards were always watching from above. This concept was employed by Foucault (1995) to discuss the role of surveillance in constructing a power dynamic between different societal groups. Remote sensing can be productively explored through this concept as it highlights the power imbalances and potential harm that can be perpetrated through its use.

Remote sensing, like a panopticon, offers unrestricted access and visibility to the analyst while being nearly invisible to those being photographed or videoed from above. Power imbalances are further skewed due to the historical dominance of availability and use of these technologies by countries and researchers in the Global North, primarily in Europe and North America (Agapiou & Lysandrou, 2015; Cohen et al., 2020; Davis, 2020). This creates a power imbalance between archaeologists and LID communities, not only in terms of surveillance dynamics, but also knowledge transfer and education when such communities know very little about remote sensing technologies and how they are being used (Davis & Douglass, 2020).

Geographers have well-established guidelines on the issues that can arise around privacy and confidentiality with the use of geospatial technologies, including remote sensing datasets. For example, the American Association of Geographers (AAG) specifically includes remote sensing within their code of ethics, stating that all decisions regarding geospatial data collection, ownership, and analysis should afford those communities “that bear the burdens of geospatial research the opportunity also to share in its benefits”. Furthermore, the AAG guidelines stipulate that field-based projects should share all results and findings with local communities and that local collaborators should be included in the dissemination of these results, including as authors on publications (AAG, 2009). There are also examples of official codes of ethics and research guidance among other fields like ethnobiology which explicitly include imagery and other research products that capture “traditional knowledge or collections of flora, fauna, or other elements of biocultural heritage found on community lands or territories” (International Society of Ethnobiology, 2006, p. 2).

Within archaeology, however, there is no such unified approach, despite the widespread and growing use of remote sensing in the discipline. This is not unique among remote sensing technologies but is true of digital methods as a whole (Dennis, 2020). For example, professional organisations like the Society for American Archaeology (SAA) have not developed specific mandates for community engagement or best practices in remote sensing/geospatial research (Davis et al., 2021; SAA, 2016). Progress has been made within the code of ethics ratified by the Computer Applications and Quantitative Methods in Archaeology (CAA) organisation. The CAA specifically acknowledges that their work, which is largely digital in scope and encompasses tools like remote sensing and geospatial technologies, can impact local communities and the general public. As such, the CAA “is committed to engagement and consultation with groups and individuals impacted by archaeological work carried out by CAA members, with the aim of building relationships that are respectful and mutually beneficial” (CAA International, 2018). While this acknowledgement is important, and there is growing consensus that engagement with local communities is needed within archaeological practice, ethical guidelines surrounding remote sensing archaeology are largely non-existent, and when present, vary widely between regions and organisations.

This is particularly troubling given the close connection between LID communities and their landscapes. In societies around the world, space is infused with cultural significance, anchoring particular places with unique cultural and social roles within communities over time (Basso, 1996; de Certeau, 1984; Ingold, 1993; Lepofsky et al., 2017). Resulting from this human place-making phenomenon, any research that surveys landscapes should be thought of in “critical terms that adhere to ethical standards related to cultural research” (Davis et al., 2021, p. 309).

This idea of culturally constructed landscapes is well established in geography and in other anthropological subfields that actively engage in human-subjects research; for example, participatory mapping studies (Álvarez Larrain & McCall, 2019; Álvarez Larrain et al., 2021; Colloredo-Mansfeld et al., 2020; Dunn, 2007; King, 2002; Ramírez-Gómez et al., 2015). One lesson these studies can teach us is that by involving local community members in analysing datasets – maps, satellite images, or otherwise – we can avoid many ethical dilemmas and negative outcomes for LID communities that result from excluding important perspectives in the research process (Álvarez Larrain & McCall, 2019; Álvarez Larrain et al., 2021; Ehrman-Solberg et al., 2020; Kwaku Kyem, 2004; Sanchez et al., 2021). While participatory research design does not alleviate all issues (see Álvarez Larrain & McCall, 2019), it sheds much needed light on ethnocentric assumptions and practices which are vital in stopping the perpetuation of colonial exploitation of marginalised communities.

A brief review of remote sensing archaeology in Africa

Aerial photographs have made a significant contribution to archaeology since their inception in the nineteenth century from hot air balloons (e.g., Capper, 1907). On the

African continent, the first published examples of aerial photographs driving archaeological research emerged in the 1950s, where researchers began to identify archaeological sites across large swathes of the landscape (e.g., Gard & Mauny, 1961; Saumagne, 1952). Over the next several decades, thousands of archaeological sites were documented using aerial imagery, including monumental architecture, urban complexes, and relationships between urban centres and hinterland settlements (Denbow, 1979; Evers, 1975; Jones, 1978; Maggs, 1976; Mason, 1968; Saumagne, 1952; Seddon, 1968).

In a 2020 article (Davis & Douglass, 2020), we identified four major trends in Africanist remote sensing archaeology: monitoring anthropogenic impacts on cultural heritage (Bewley et al., 2016; Casana & Laugier, 2017; Hobson, 2019; Lasaponara & Masini, 2018; Parcak, 2007; Parcak et al., 2016; Parcak, 2009), the use of vegetation indices for identifying archaeological deposits, geomorphological mapping to identify relationships of past landscapes and ancient settlement patterns, and the analysis of remote sensing data using semi- and fully-automated means (i.e., machine learning, AI). Since the publication of that article, many of these trends have continued, but there has also been an increase in recent applications of remote sensing across the continent. With such an increase, a closer look at how these studies are conducted is warranted to ensure they are used ethically.

In Eastern Africa, researchers in Ethiopia have demonstrated the role that remote sensing can play in preserving cultural heritage sites (Ochungo et al., 2022). Additionally, Tanzania has seen an increase in remote sensing archaeological investigations. Fitton et al. (2022) conducted a study in Zanzibar and demonstrated how remote sensing, when combined with pre-existing archaeological datasets, can provide important insight into the historic change of landscapes and conditions at archaeological sites in the present (see also Alders, 2023). This information can be invaluable for archaeological prospection and characterising archaeological landscapes. Sadasivuni and colleagues (2022) also used remote sensing datasets to examine settlement and mobility choices as adaptive strategies for climate change. A common thread in most of these studies is their focus on methodological development and serving to leverage remote sensing for academic aims. However, these studies do not explicitly discuss any participatory involvement of local community members or larger considerations of research ethics or approval. This is not to say these studies were improper, but rather speak to a wider global issue in the field where such considerations are rare.

These considerations can become incredibly important when dealing with sensitive subjects. One area of research in Africa, where there has been an increase in recent remote sensing studies, is that of funerary landscapes. Funerary landscapes can be defined as a “specific type of archaeological landscape revealing the phenomenological relation between death, disposal of the body in the environment and the social memory of the group participating in the remembrance of the burial” (Daróczi, 2012, p. 202). In Sudan, for example, Costanzo et al. (2021) use open-source remote sensing datasets from QuickMapServices to map the locations of thousands of funerary monuments and features over an area of over 4100 km². To analyse the spatial patterns among these landscape features, the authors turn to point pattern analysis, which

is a form of spatial regression that can be applied to examine relationships between individual archaeological features and between archaeological and environmental variables (e.g., Davis et al., 2020b, 2020c). Costanzo and colleagues demonstrate that funerary monuments were drastically underestimated in prior research and show how remote sensing and geospatial analysis can shed light on the scale and organisation of archaeological landscapes. However, the use of publicly available datasets collected by agencies outside of Sudan to record highly ceremonial and culturally significant features does pose potential issues. In this instance, the authors worked with local archaeological authorities and ministries, clearly following all legal requirements to conduct this work, but the degree of involvement (if any) with local communities themselves is not specified.

Further studies of funerary landscapes come from Western Africa. For example, researchers in Mauritania have demonstrated the importance of open-source remote sensing datasets from Bing Maps to identify archaeological features and funerary landscapes (Linares Matás & Lim, 2021). Additionally, the researchers make use of a suite of geospatial analyses (including clustering and viewshed analyses) to infer relationships between different features on the landscape. Building on this work, follow-up research has used the features identified by Linares Matás and Lim (2021) to examine the social and economic dynamics of labour mobilisation to construct funerary monuments in Mauritania (Linares Matás, 2023). Importantly, Linares Matás and Lim (2021) devote a considerable portion of their article to discussing the ethical implications of their work, paying particular attention to the need for more community engagement and participation within studies of landscapes. The authors even develop a framework for informing remote sensing archaeology that ensures local community involvement in the research process. This serves as a key example of how participatory research can be encouraged and improved within Africanist remote sensing archaeology.

Finally, there has been an increase in machine learning approaches to archaeological remote sensing within Africa. For example, Bachagha et al. (2023) use a random forest algorithm to identify fortified Roman sites in Tunisia. Also, Davis and Douglass (2021) use a similar approach to identify foraging settlements on Madagascar. Machine learning raises additional ethical issues that go beyond the scope of this chapter but must also be critically addressed in close collaboration with stakeholder communities (see Eubanks, 2017; Huggett, 2021). As we describe below, an important initial step in any research agenda (including those involving machine learning and AI) is examining the potential negative consequences of the work and discussing these concerns with those who are potentially at risk of harm.

Altogether, there has been a surge in remote sensing archaeological achievements in Africa over the past several years alone, with emphasis on both prospection for archaeological sites and extensive geospatial analysis of patterns that emerge from these remotely acquired datasets. Additionally, several studies are making use of topographic information from digital elevation models, which open up exciting new avenues for research looking at landscape modifications, including within heavily

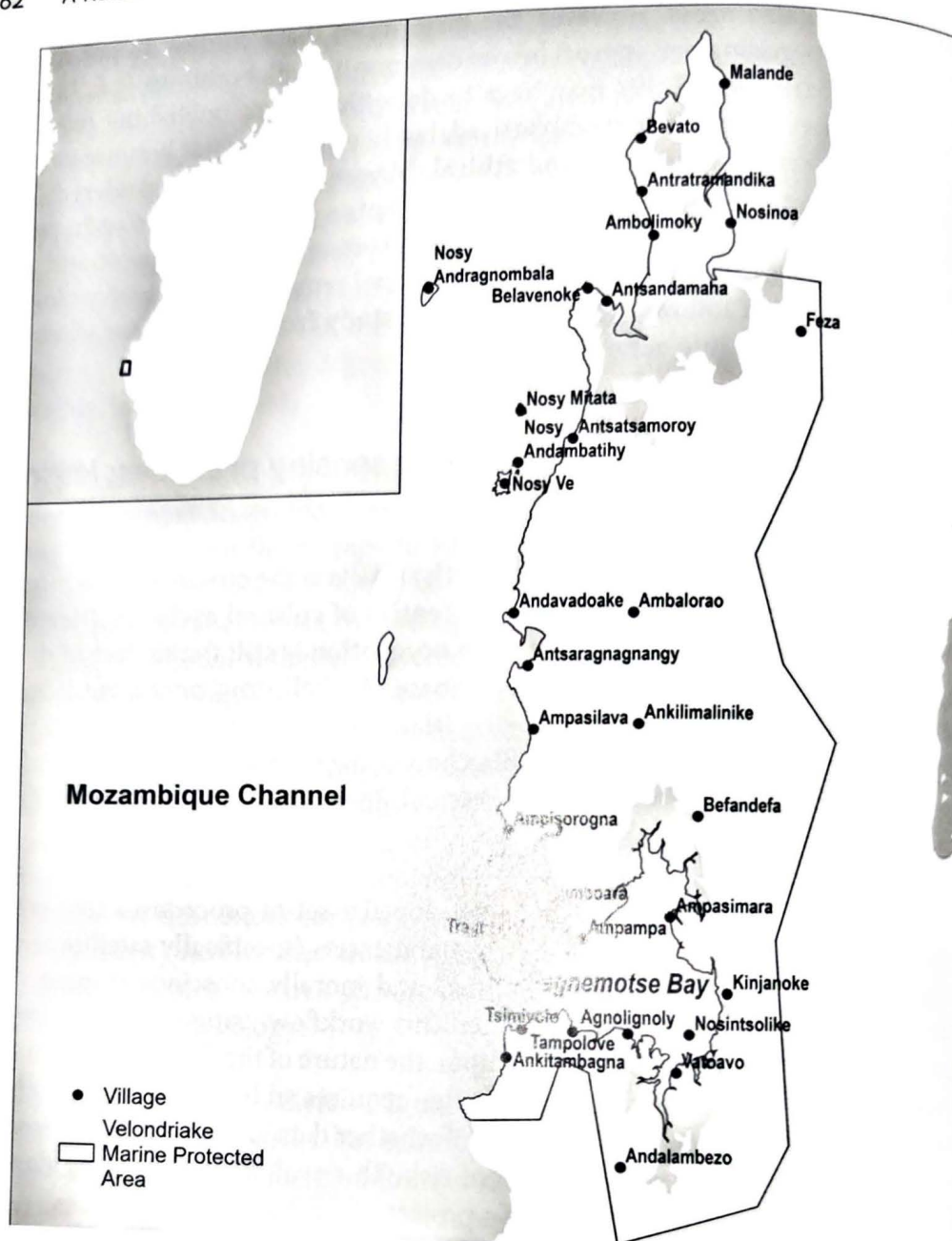
forested and vegetated areas. However, the majority of these studies do not explicitly engage with participatory approaches or community collaborations (c.f., Linares Matás & Lim 2021). Part of this may have to do with current publishing practices where collaborative efforts are not emphasised, but in other cases it is because participatory methods were not involved, and ethical dilemmas were not considered. This lack of attention is not limited to Africa but represents a global issue. As such, one of the goals of this chapter is to highlight ways that community engagement and participatory practices can be integrated into traditional remote sensing archaeological investigations. In what follows, we present a case study from Madagascar where we put many of these ideas into action.

Towards ethical archaeological remote sensing practices: lessons from fieldwork on Madagascar

Madagascar is the fourth largest island on Earth and sits at the crossroads of a number of historically important trading routes and centres of cultural exchange (Beaujard, 2007). The timing of the island's first human occupation is still the subject of debate (Douglass et al., 2019; Hansford et al., 2018, 2020; Mitchell, 2019, 2020), but it is well established that humans have lived on the island for at least the past 2000 years (Douglass et al., 2019). The authors of this chapter have, collectively, been conducting remote sensing-based archaeological research in southwest Madagascar for over a decade (Figure 4.1).

Based on our own experiences, we have developed a set of procedures that we try to follow to ensure our studies using geospatial datasets (specifically satellite remote sensing imagery) are conducted in an ethical and morally conscious manner. This is illustrated in Figure 4.2. We implemented this workflow using primarily satellite imagery in rural Madagascar. Depending upon the nature of the data being evaluated, these steps may change. The use of drones often requires additional legal regulations to be adhered to by researchers. Regardless of whether data already exist, all research projects should begin with an evaluation of risk. This evaluation is not singular but should be repeated throughout the life of a project.

In brief, for any study involving remote sensing data, the first step a researcher must undertake is a risk assessment: what is the potential that this analysis causes people harm and how likely is that negative outcome? In some instances, negative impacts are obvious (i.e., trying to identify locations that local communities want kept secret). In other instances, negative impacts are not as clear; ultimately, they will require LID community input. As such, it is always advised to consult with community members, but when negative consequences are clear this consultation is required. After this initial risk assessment, it is important, particularly for research where new data are being collected, that a data management plan is established. Ideally, this should be conducted with local stakeholders to ensure that they have control over how (and if) data about their territory are shared and presented to the public. In what follows,

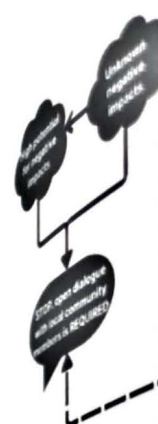


Source: Esri, TomTom, Garmin, FAO, NOAA, Esri, © OpenStreetMap contributors, TomTom, Garmin, Foursquare, METI/NASA, USGS.

Figure 4.1 Map of the study region on Madagascar

we discuss some of the lessons we have learned in working in collaboration with LID community members when planning and implementing these projects.

Madagascar has ratified several international treaties that relate to space-based communication and exploration technology (Oduntan, 2019), but there are no statutes that require prior consent or consultation with LID communities when accessing or

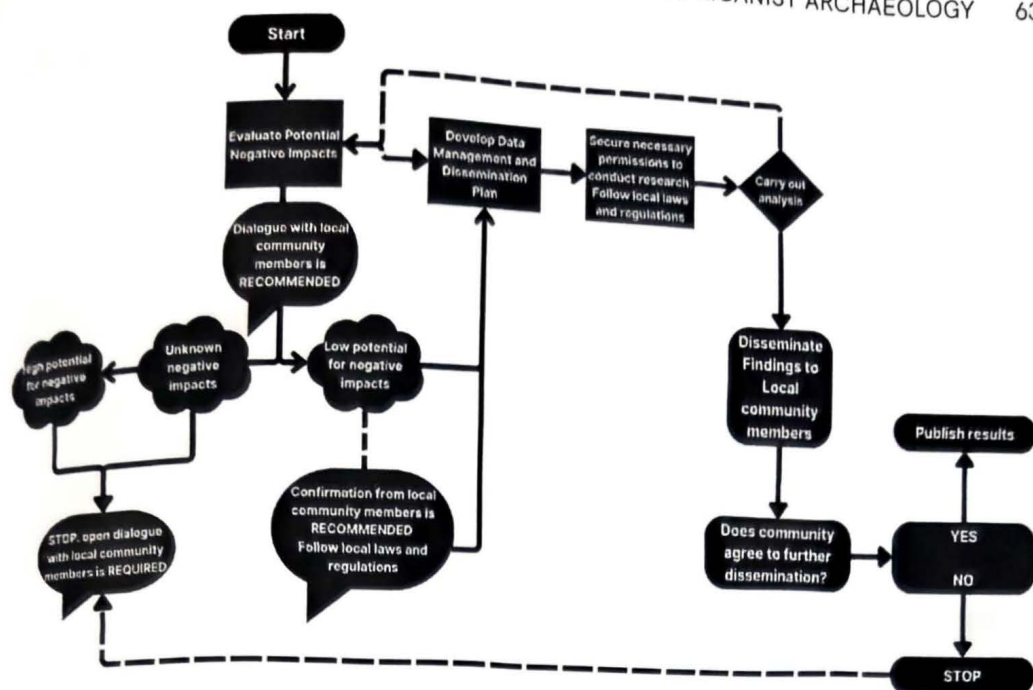


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Figure 4.2 Workflow of ethical aerial remote sensing research procedure

commissioning spaceborne images of Madagascar's territories (Davis et al., 2021). In contrast, uncrewed aerial vehicles (UAVs) or drones are more closely regulated (see "Instruction N°01 ACM/DRG/17 relative aux conditions d'exploitation des aéronefs télépilotes", <https://www.acm.mg/>). Legal codes stipulate that drones cannot be flown within 100 metres of an individual unless prior consent is obtained from those who may be recorded, and drones are expressly prohibited from flying over areas where terrestrial access is restricted. The difference in legal regulation between remote sensing technologies and data raises questions concerning what LID communities are aware of with respect to data acquisition, how communities feel about being monitored from space without consent, and what, if any, steps we as researchers should be taking when accessing and analysing these kinds of datasets (see, for example, Vargas-Ramírez & Paneque-Gálvez, 2019).

One of the present authors (KD) established a community archaeology organisation, the Morombe Archaeological Project (MAP) in 2011. Since this time, the MAP team has increasingly incorporated remote sensing methods like UAVs into field projects. When the present author (DSD) began his dissertation project in 2018, the introduction of satellite-based remote sensing methods opened new avenues for research, but also new sets of challenges to ensure equitable community collaboration and consent. Over the course of many extensive conversations with MAP and LID community members in southwest Madagascar, and observations during surveys, we found that in many instances, photographing places was not of major concern to most individuals (for example, photographing village boundaries and houses; Davis et al., 2021).

However, targeting sacred places (like ancestral burials and tombs) was in violation of local taboos (*fady*) and against the wishes of LID community members.

During DSD's dissertation fieldwork, there were instances when areas identified in satellite imagery corresponded with sacred spaces. In some instances, this was known prior to fieldwork, and these places were automatically identified by LID community members and avoided by the archaeological team. In other instances, it was not clear until we were conducting field investigation that a location was sacred. In these instances, we avoided *fady* places by re-routing our surveys to circumvent these locations. Such outcomes were only achievable because the MAP team is comprised of LID community members who could point these locations out and ensure we adhered to LID customs and wishes.

The surveys described above resulted in a significant level of discovery, documenting hundreds of archaeological deposits along the southwest coastline and generating an important dataset that has since shed light on settlement and mobility decision-making, social networks, technology, and long-term human–environment interactions (Davis & Douglass, 2021; Davis et al., 2020a, 2020b, 2023a, 2023b). At the conclusion of this project, all the results from these surveys have been stored within databases that are kept by MAP and are also backed up on additional hard drives at Columbia University with local community permission. This is not publicly shared (to protect sensitive data) but serves as a backup to the digital files kept on Madagascar. While the results of such studies are significant for Malagasy archaeology and global archaeological studies of human landscape dynamics, what is more important is how the work was conducted and what progress can be made to make future work even more inclusive and equitable.

Our experiences in conducting remote sensing archaeology on Madagascar also highlight a fundamental problem with this kind of research when ethics are not considered: data access and ownership (e.g., Gupta et al., 2020; Madsen, 1994; Reddix-Small, 2014). Unlike UAVs, where permits and legal statutes must be obtained and followed, our satellite-based research required no such permissions and could have resulted in immense cultural transgressions (i.e., disturbance and/or desecration of ancestral remains) that would have deeply hurt local communities. Participatory approaches are, therefore, vital to the conducting of just and equitable scientific practices. This process requires collaboration across all dimensions of remote sensing research, extending to project conceptualisation, field verification of remotely sensed information, and data access and ownership.

When thinking about satellite imagery specifically, our experiences have resulted in some examples of basic principles that support the ethical and equitable use of remote sensing technology in archaeology. These principles centre on collaboration and co-production of knowledge throughout the research process, beginning with forming research questions and continuing to research design and planned outcomes of projects (see also Derickson & Routledge, 2015). First, researchers can and should use remote sensing tools, as they are often foundational to collecting preliminary data

and forming research questions; however, we must use these datasets with the understanding that LID communities retain the right to say no to any investigations of ideas or preliminary observations made from remotely sensed data. Second, researchers must be transparent about where satellite imagery comes from and convey this to the communities we work with. Third, researchers must ensure a clear data management and sharing plan for any information collected using remotely sensed information. While satellite images may be freely available, the data we derive from these sources are not, and some communities may wish to keep information private. By adhering to these principles, we can spur on new and exciting research while ensuring local communities have autonomy on how this work is carried out.

Summary

Remote sensing is now a standard method used within archaeological investigations. The use of remote sensing archaeology has been rapidly expanding, and within Africa, exciting trends in archaeological discovery are being facilitated by the use of these datasets. With any technological advancement, however, we must be attuned to the different repercussions that such methods may have, particularly with regard to community stakeholders.

As we have argued here, remote sensing can exacerbate inequalities and power asymmetries between researchers and community stakeholders in archaeological landscapes. Alleviating these issues requires collaborative research practices that involve community stakeholders in the scientific process. In this way, data sharing, the exact objectives of research programmes, and guidelines for how certain technologies should be used can be established collaboratively and in a way that reduces power imbalances and mitigates potential harm. By engaging in these important discussions, we can limit the potential for research projects to violate LID community autonomy, sovereignty, traditional customs, and privacy. As we advocate elsewhere, archaeological projects incorporating remote sensing should strive to disseminate information about how the remote sensing technology used within a project works, including by providing training in how to use these tools, and ensure that the data acquired are available to LID communities and according to a plan developed in collaboration with stakeholders (Davis et al., 2021).

A brief guide to recommended reading

There has been a recent uptick in awareness by archaeologists about the ethical dilemmas associated with remotely sensed datasets and geophysical data collection and analysis. Nevertheless, at the time of writing this chapter, there is no overarching book or chapter that solidly encapsulates the current guidelines or best practices. For those interested in reading further on this topic, we highlight a few seminal references.

There are currently a handful of special issues that are dedicated to the ethics of remote sensing in archaeology and the use of remote sensing in collaboration with indigenous communities (Davis & Sanger, 2021; Sanger & Barnett, 2021; Sturm & Herrmann, 2024). Within each of these special issues are dozens of important articles that provide important examples of how remote sensing studies can be carried out in an inclusive and ethical manner, spanning global contexts.

Outside of special issues, there are other important articles that provide some consensus on best practices for how to approach remote sensing investigations in a way that is inclusive and engaged with local community concerns. This includes calls for ethical guidelines and decolonised approaches (Chase et al., 2020; Cohen et al., 2020; Dunnivant et al., 2023), new frameworks for participatory geospatial research (Gleason et al., 2023), as well as exposés on how researchers have tackled contentious issues or sensitive subjects during fieldwork (González-Tennant & González-Tennant, 2023). Between these aforementioned articles and special issues and the rest of this chapter's list of references, readers will have a good starting point for delving more deeply into this topic.

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